Pathlavath Kumar

em25mtech14010@iith.ac.in | kumarpathlavath15@gmail.com | +91 8104091340 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
SANKALP JUNIOR KALASALA	Class XII (Telangana State Board of Intermediate Education)	2020	97.80%
Z P HIGH SCHOOL KULKACHERLA VIKARABAD	Class X (Board of Secondary Education Telangana)	2018	8.80%

PROFESSIONAL SUMMARY

M.Tech student with project experience in Python, SQL and RAG, analyzing technology, market and policy data to identify evidence-based R&D opportunities. Experience in cross-cultural collaboration with Japanese startup founders and university students.

EXPERIENCE

Wadhwani School of Data Science and AI (WSAI), IIT Madras | Research Internship

- Conducted **AI research and data analysis** using Python, Pandas and NumPy to structure **academic, interest and preference data**, supporting systematic analysis and **evidence based recommendations** across datasets.
- Built **RAG pipelines and fine tuned LLMs** using domain datasets and LLM APIs to **structure, retrieve and validate source grounded information**, improving contextual and knowledge based career guidance workflows.
- **Analyzed and validated AI generated outputs** for accuracy, relevance and reliability, identifying **response patterns and information quality issues** to improve knowledge retrieval and recommendation workflows

Tech Mahindra | Business Development (Trainee Associate)

- Conducted **technology and business research** for the **Accu Chek product**, analyzing requirements, technology gaps, customer needs and process improvement areas to structure relevant information and identify potential opportunities for **technology driven business improvement**.
- Collaborated with **cross functional teams and stakeholders** to understand Accu Chek requirements, evaluate technology solutions and organize findings into **actionable insights**, supporting solution assessment, problem solving and informed business decisions.
- Synthesized and communicated **product, technology and project insights** by consolidating research findings, solution evaluations and key requirements into clear outputs that supported **opportunity identification, stakeholder discussions and data driven decision making**.

PROJECTS

ChargeIndia | EV Adoption & Charging Demand Intelligence for India (GitHub)

Python, Pandas, NumPy, Scikit learn, HistGradientBoosting, SQL, Streamlit

- Engineered an **EV market intelligence pipeline** processing **11,250+ VAHAN records across 30+ States/UTs**, integrating EV adoption, charging infrastructure and policy signals with data quality and provenance tracking.
- Developed a **HistGradientBoosting time series forecasting model** using lag, rolling statistics and growth features with chronological validation, achieving **4.58% WAPE and -0.13% forecast bias**, reducing error by **87.12% versus baseline**.
- Built **Potential Charging Demand Index and R&D opportunity screening models** to identify regional EV opportunities from adoption and infrastructure gaps, identifying **Telangana/Hyderabad for EV WPT demonstration screening** and presenting findings through an interactive Streamlit dashboard.

AgentHire | Agentic AI Recruitment Platform (GitHub)

LangGraph, RAG, Qdrant, BAAI Embeddings, Next.js, FastAPI

- Designed a **7 agent LangGraph workflow** to process **unstructured candidate data**, extract requirements and orchestrate automated evaluation across multiple AI stages.
- Built a **RAG based semantic matching engine** using **Qdrant and BAAI embeddings** to connect candidate capabilities with structured job requirements through **context aware retrieval and similarity matching**.
- Implemented **human in the loop validation checkpoints** and traceable workflow states to improve **AI reliability, decision transparency and controlled evaluation**.

TestForge AI: AI Powered Test Case Generation Platform (GitHub)

- Built a **full stack** developer tool (**React, Vite, Tailwind, Node.js, Express, MongoDB**) that scans repositories and uses **LLM analysis** to auto generate unit, integration, API and edge case tests.
- Used **Spec Driven Development**, writing a complete technical specification and directing an **AI coding agent (GitHub Copilot/Codex)** to implement the platform phase by phase.
- Built a **memory embedding layer** in MongoDB that retrieves similar past generations to improve future test quality.
- Implemented **JWT authentication**, password hashing, upload sanitization and **prompt injection defenses**, with an approve/reject/regenerate feedback loop to validate AI outputs.

SKILLS

Programming Languages: C++, Python, SQL

Technologies: MongoDB, Redis, Qdrant, Docker, Git/GitHub, GitHub Actions

Frameworks & Libraries: Next.js, FastAPI, Node.js, Express.js, React, Pandas, NumPy, PyTest, Jest

AI/ML: LLMs, Generative AI, RAG, AI Agents, LLM APIs, Prompt Engineering, Embeddings

JAPAN ENTREPRENEURSHIP & CROSS CULTURAL COLLABORATION

- Worked with **Japanese startup founders** to refine venture ideas, analyze market opportunities, develop the final concept, and pitch the solution through **cross cultural collaboration**, securing **3rd place in the Tongali Project**.
- Collaborated with **Shimane University students from Japan** to develop solutions to real world problems.

SCHOLASTIC ACHIEVEMENTS

- Ranked among the **Top 1%** of 7,000+ applicants in the **national selection** process for the WSAI, **IIT Madras** Internship.
- **Secured 97.80% in Class XII, achieving College Topper and State 4th Rank in the Telangana State Board of Intermediate Education.**

POSITIONS OF RESPONSIBILITY

- **Head, Operations & Logistics, MSME Tech Connect,** IIT Hyderabad**:** Led **100+ exhibitors, 40+ investors & 2,000+ visitors**, coordinating stakeholders, communication and event operations for a **Govt. of India initiative**.
- **Assistant Manager, E Cell,** IIT Hyderabad**:** Engaged **startups, founders, investors, VCs & alumni**.